
RISKCOURT

An AI prediction market for options alpha

Independent jurors form calibrated odds. Deterministic policy trades only when jury odds beat the option-implied hurdle.

ARGUE

CALIBRATE

GATE

ACT

Hackathon technical overview

Recorded-first • Alpaca paper mode • No live-money path

AI forecasts are cheap; disciplined entry is not

A model can be persuasive and still be wrong, stale, oversized, or impossible to audit.



Jury odds only matter when they clear the hurdle

The option quote supplies a transparent break-even proxy; the risk kernel decides whether an edge is tradable.

THE FORMULA

$$P_{\text{hurdle}} = (D + S) / W$$

D net debit per share

S slippage buffer

W strike width

RECORDED SPY EXAMPLE

57.4%

calibrated jury probability

32.0%

option-implied hurdle

SPY \$5-wide call spread • \$1.50 debit • \$0.10 buffer

Edge = 25.4 percentage points • minimum = 8 points

Payoff geometry is a proxy—not a literal physical probability.

Every path can end in abstain

Model confidence is never a substitute for fresh evidence, bounded risk, or an explicit approval.



Maximum loss = debit $\times$ 100 + fees

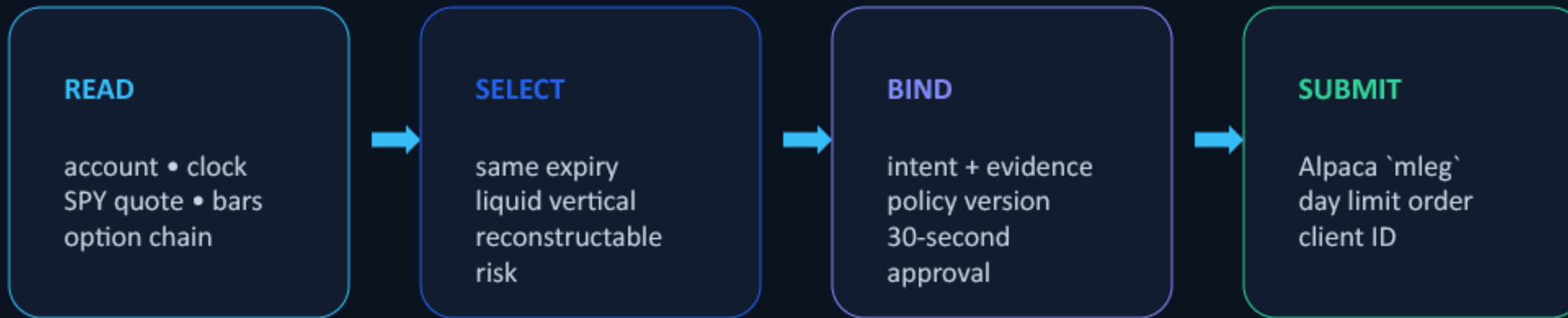
One contract • 0.5% of account equity cap

Fail closed

stale • missing • disagreement • duplicate • kill switch

A paper order is the last step, not the first

Alpaca reads, approval binding, and idempotent execution keep the AI boundary narrow.



After submit: accepted → partial/filled → canceled/rejected → exit → P&L snapshot

The official MCP/CLI proof and a market-open paper order are still release gates.

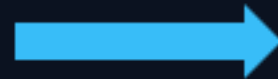
The story is replayable before the market is open

Recorded cases exercise the same decision boundaries without credentials, network access, or fake live results.

12 CASES

trade • abstain
veto • error

stale quotes • illiquidity
no edge • disagreement
injection • outage • duplicate



JUDGE-FACING DECISION CARD

jury odds → hurdle → edge
selected legs → max loss
order lifecycle → paper P&L

Five consecutive golden rehearsals are timestamped in the repository.

What is ready—and what still needs the market

RiskCourt is honest about the difference between a verified local release candidate and a judged deployment.

READY LOCALLY

- ✓ deterministic probability + hurdle
- ✓ bounded multi-leg paper mapping
- ✓ evidence-bound approvals
- ✓ replay, evaluation, browser, security gates

OPEN RELEASE GATES

- official Alpaca MCP/CLI proof
- fresh-account market-open order
- terminal lifecycle + P&L read
- hosted verification and final LabLab fields

No live-money execution path exists. Paper performance is not a promise of future returns.